MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

TRANSFORMER, SCOTT-T, ANALOG, KEY CODE PTE

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic Scott-T transformer module.

1.2 Classification. This specification covers one class of modules.

Class I as specificed in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.
MIL-T-27 - Transformers and Inductors (Audio, Power, and High-Power Pulse), General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein. Each module transformer shall comply with the requirements of MIL-T-27 type TF5RX21ZZ.

3.2 Correlation requirments . Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.076 failures per million operating hours when predicted in accordance with MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be PTE.

3.5.2 Size. The module size shall be 1B.

3.5.3 Weight. The module shall have a maximum weight of 80 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A, and heat dissipation fins on increments A and B.

3.5.6 Depth. The module shall have no additional depth requirements, except the only parts of the module permitted to extend below the interface plane shall be the keying pins, contact pins, and pin shields.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

* 3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/226-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film unless otherwise specified. All diodes shall be JAN TX 1N4465 or equivalent.

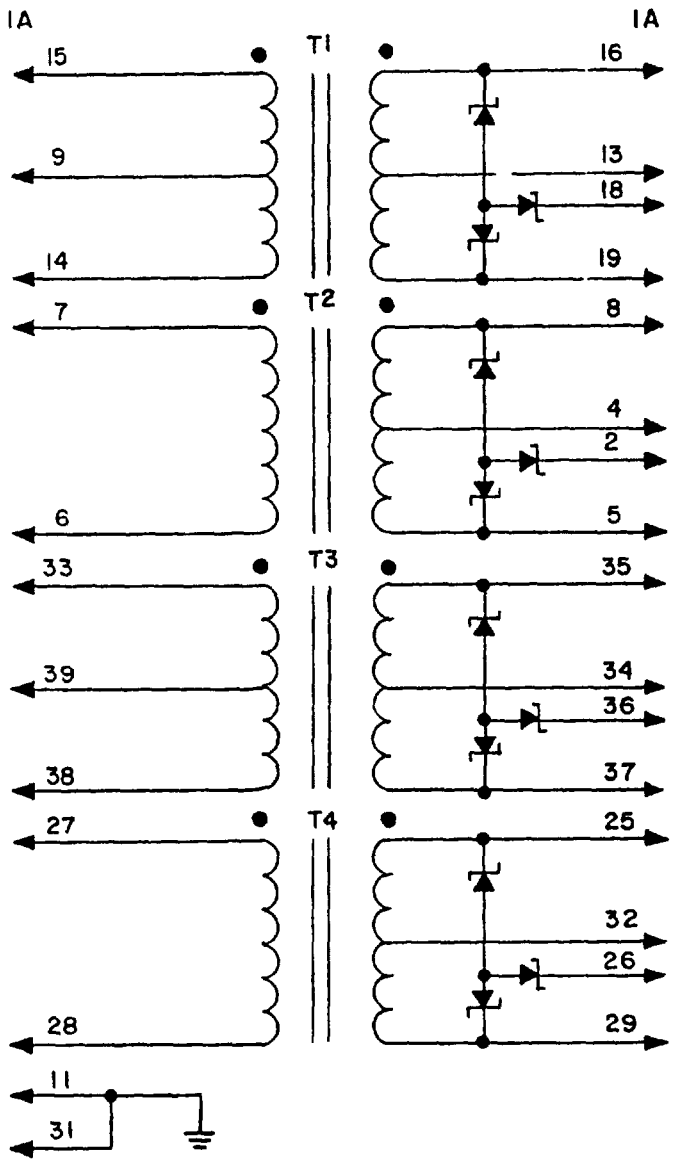
4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through N shall be performed in any order with the following exceptions. If test condition A is part of a sequence of tests, the order of testing shall be as follows: A where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

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NOTES:
1. Pins 1A-1, 3, 10, 12, 17, 20, 21, 22, 23, 24, 30, and 40 are unused.
2. All diodes shall be JAN TX 1N4465 or equivalent.

FIGURE 1. PTE module functional diagram, Scott-T transformer.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/- 3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A5	Isolation	II	X	10	∞	1	∞	10	∞	1	∞	MΩ
B1	Continuity	III	X	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	Ω
B2	Continuity	III	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	
C1-C2	DC resistance	IV	S	150	400	150	400	135	480	135	480	
C3-C4	DC resistance	IV	S	75	250	75	250	65	420	65	420	
C5-C8	DC resistance	IV	S	4.0	10.0	4.0	10.0	3.5	12.0	3.5	12.0	Ω
D1-D4	Voltage suppression	V	Q	16.0	26.0	16.0	26.0	16.0	26.0	16.0	26.0	V_{p-p}
D5-D12	Voltage suppression	V	X	16.0	26.0	16.0	26.0	16.0	26.0	16.0	26.0	V_{p-p}
E1-E2	Polarity	VI	X	3.2	3.8	-	-	-	-	-	-	V
F1-F3	Angular accuracy	VII	X	-0.020	0.020	-0.020	0.020	-0.025	0.025	-0.025	0.025	deg
F4-F11			X	-0.040	0.040	-0.040	0.040	-0.045	0.045	-0.045	0.045	
F12-F14			X	-0.020	0.020	-0.020	0.020	-0.025	0.025	-0.025	0.025	
F15-F22			X	-0.040	0.040	-0.040	0.040	-0.045	0.045	-0.045	0.045	
F23			X	-0.020	0.020	-0.020	0.020	-0.025	0.025	-0.025	0.025	
F24-F31			S	-0.040	0.040	-0.040	0.040	-0.045	0.045	-0.045	0.045	
F32			X	-0.020	0.020	-0.020	0.020	-0.025	0.025	-0.025	0.025	
F33-F40			S	-0.040	0.040	-0.040	0.040	-0.045	0.045	-0.045	0.045	
F41-F80	Angular accuracy	VII	S	-0.070	0.070	-0.070	0.070	-0.070	0.070	-.080	0.080	deg

See footnotes at end of table.

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TABLE I. Module electrical requirements -Continued.

Test condition	Parameter	Ref test table	Test type 1/- 3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
G1-G22	Differential quadrature	VIII	S	-3.50	3.50	-3.50	3.50	-4.00	4.00	-4.00	4.00	mV
G23-G40			S	-1.50	1.50	-1.50	1.50	-1.75	1.75	-1.75	1.75	
G41-G62			S	-3.75	3.75	-3.75	3.75	-4.25	4.25	-4.25	4.25	
G63-G80	Differential quadrature	VIII	S	-1.75	1.75	-1.75	1.75	-2.00	2.00	-2.00	2.00	mV
H1-H4	Common mode rejection ratio	IX	Q	-94	-∞	-94	-∞	-94	-∞	-94	-∞	db
H5-H8	Common mode rejection ratio	IX	Q	-68	-∞	-68	-∞	-68	-∞	-68	-∞	db
J1-J2	Transformation ratio	X	Q	.03850	.03928	.03850	.03928	.03850	.03928	.03850	.03928	V/V
J3-J4	Transformation ratio	X	Q	.01100	.01122	.01100	.01122	.01100	.01122	.01100	.01122	V/V
K1-K2	Phase shift	XI	Q	-0.1	0.1	-0.1	0.1	-0.1	0.1	-0.1	0.1	deg
K3-K4	Phase shift	XI	Q	-0.2	0.2	-0.2	0.2	-0.2	0.2	-0.2	0.2	deg
L1-L4	Load regulation	XII	Q	0.005	0.100	0.005	0.100	0.005	0.100	0.005	0.100	%
M1-M4	Input impedance	XIII	X	100	300	90	350	85	450	80	500	kΩ
M5-M6	Input impedance balance		X	0	20	-	-	-	-	-	-	%
* M7-M10	Input impedance angle		Q	5	85	5	85	5	85	5	85	deg
M11-M14	Input impedance		S	100	450	90	500	85	500	80	550	kΩ
* M15-M18	Input impedance angle	XIII	Q	5	50	5	50	5	50	5	50	deg
N1-N12	Crosstalk	XIV	Q	0.0	0.2	0.0	0.2	0.0	0.2	0.0	0.2	mV

NOTES:

1/ "X" Parameters shall be 100 percent tested during module acceptance tests.
2/ "S" Parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" Parameters shall be tested during qualification testing only.

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4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through N in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions A and C through N shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be nonoperating. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests A and C through N shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A through N shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Part number M28787/226-1 supersedes part number M28787/226. The two parts are interchangeable.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Frequency	390 to 820 Hz
Dielectric withstanding voltage	500 ±50 Vdc
Synchro voltage	90 Vrms line-to-line
Transformation ratio	0.03889 ±0.00039 (90:3.5) 0.01111 ±0.00011 (90:1.0)
Regulation	0.1% from no load to 50 kΩ resistive load
Phase shift	±0.1 degree (50 kΩ load)
Input impedance (V_I = 90 Vrms line to line)	80 kΩ minimum at 400 Hz 100 kΩ minimum at 800 Hz
Angular error	@400 Hz ±0.045 degrees maximum @800 Hz ±0.080 degrees maximum

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Clamping circuit voltage	26 V_{p-p} maximum
Quandrature 400 Hz full winding tap	+4.0 mV maximum ∓1.75 mV maximum
800 Hz full winding tap	+4.25 mV maximum ∓2.0 mV maximum

6.5 Module function. This module contains two 390 Hz to 820 Hz Scott-T transformers which convert , by external pin programming, 90 Vrms synchro signals to 3.5 Vrms and 1.0 Vrms resolver signals. Each secondary winding has a suppression circuit to limit output voltage transients to 26 V_{p-p} maximum.

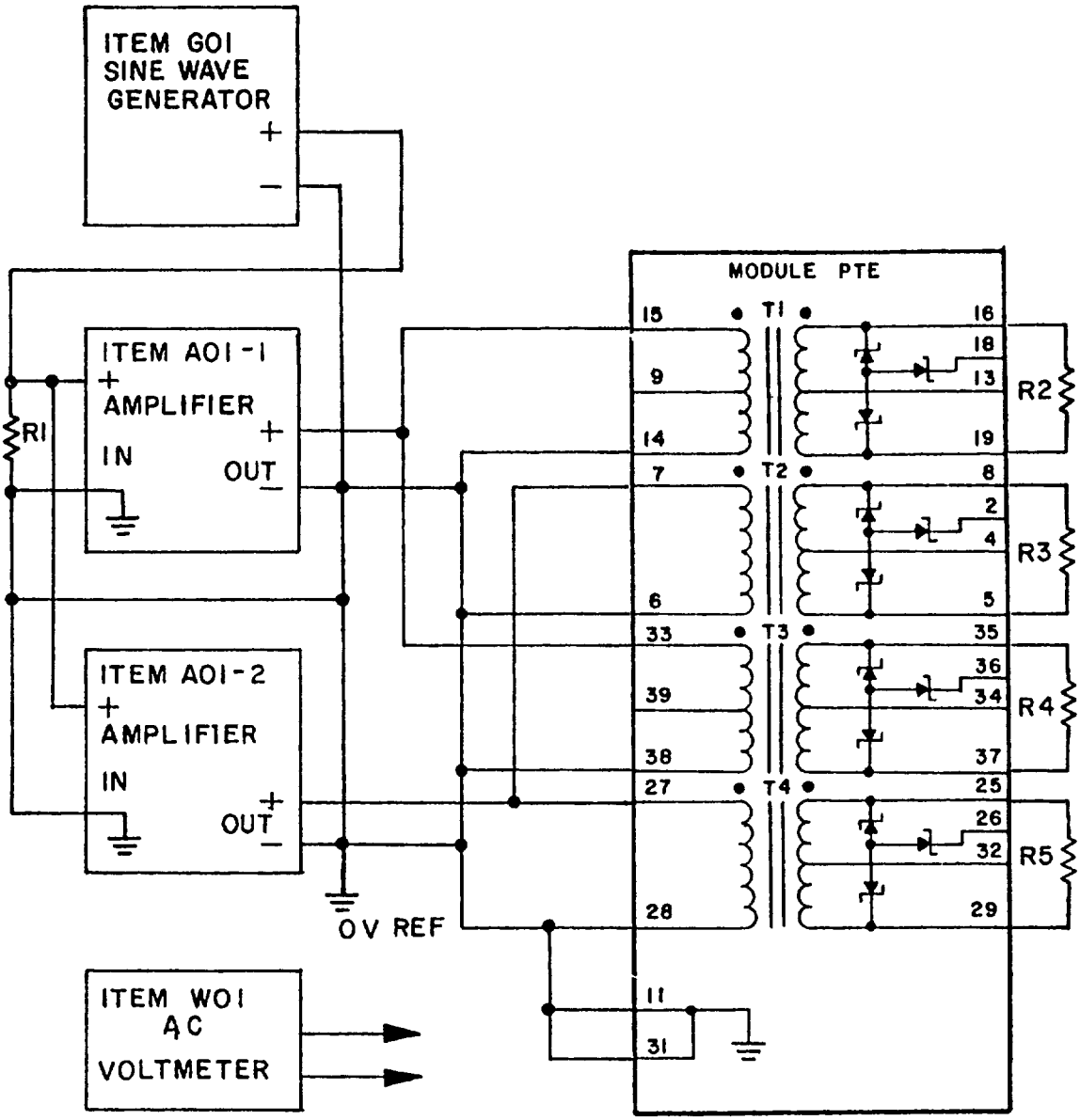
6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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NOTES:
1. R1 = 51Ω ±5%, 1/8 W.
2. R2-R5 = 50 kΩ ±1%, 1/8 W.
3. Adjust items A01-1 and G01 for an output voltage of 90 ±1 Vrms at 430 ±40 Hz at the output of item A01-1.
4. Adjust item A01-2 for an output voltage of 78 ±1 Vrms at the output of item A01-2.

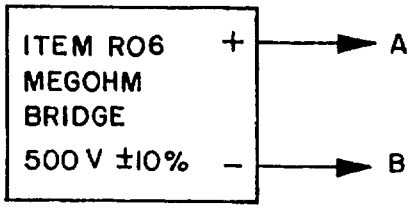
FIGURE 2. Operating temperature, component temperature and life test circuit.

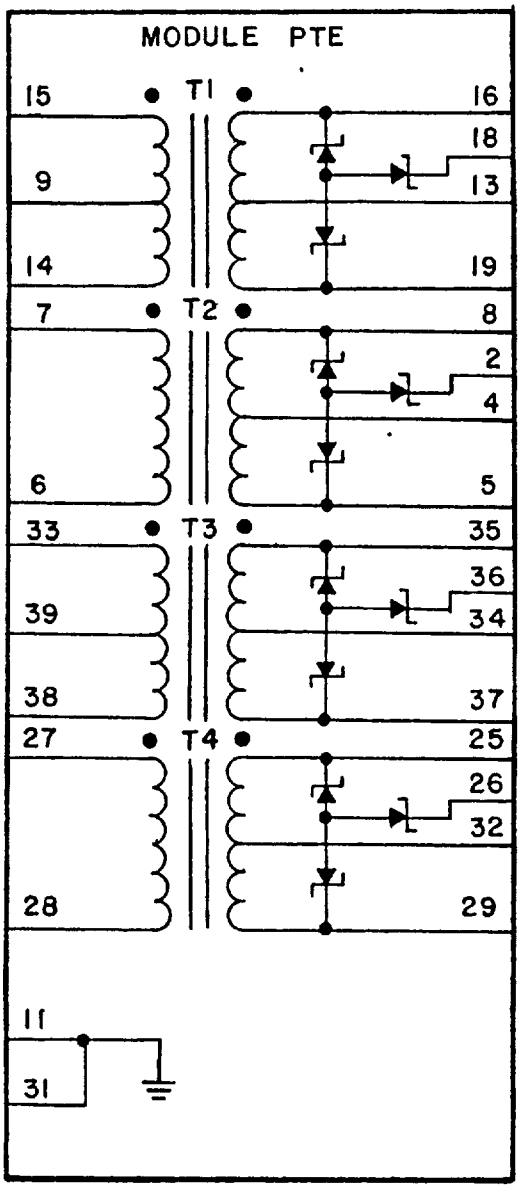
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TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	1,3,11,17, 20,21,22, 23,24	4,6,9,10, 12,13,27, 30,32,34, 39,40	10	∞	MΩ	3
A2		1,3,10,12, 17,20,21, 22,23,24, 30,40	4,6,9,11, 13,27,32, 34, 39				
A3		1,3,4,6,9, 10,12,13, 17,20	21,22,23, 24,27,30, 32,34,39, 40				
A4		1,3,4,6,10, 21,22,27, 30,32	9,12,13, 17,20,23, 24,34, 39, 40				
A5	Isolation	1,4,13,17, 21,23,32, 34	3,6,9,20, 22,24,27, 39	10	∞	MΩ	3

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge at "discharge" before connecting or disconnecting the test leads.

FIGURE 3. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
B1	Continuity	11	31	Open	Open	0.0	1.0	Ω	4
B2	Continuity	11	Module frame 1/	Open	Open	0.0	1.0	Ω	4

1/ Penetrate the anodized finish of the module frame with a sharply pointed probe. Connect the negative lead of the ohmmeter to the probe.

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TABLE IV. DC resistance test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
C1	DC resistance	Open	Open	15	14	150	400	Ω	4
C2				33	38	150	400		
C3				7	6	75	250		
C4				27	28	75	250		
C5				16	19	4.0	10.0		
C6				35	37	4.0	10.0		
C7				8	5	4.0	10.0		
C8	DC resistance	Open	Open	25	29	4.0	10.0	Ω	4

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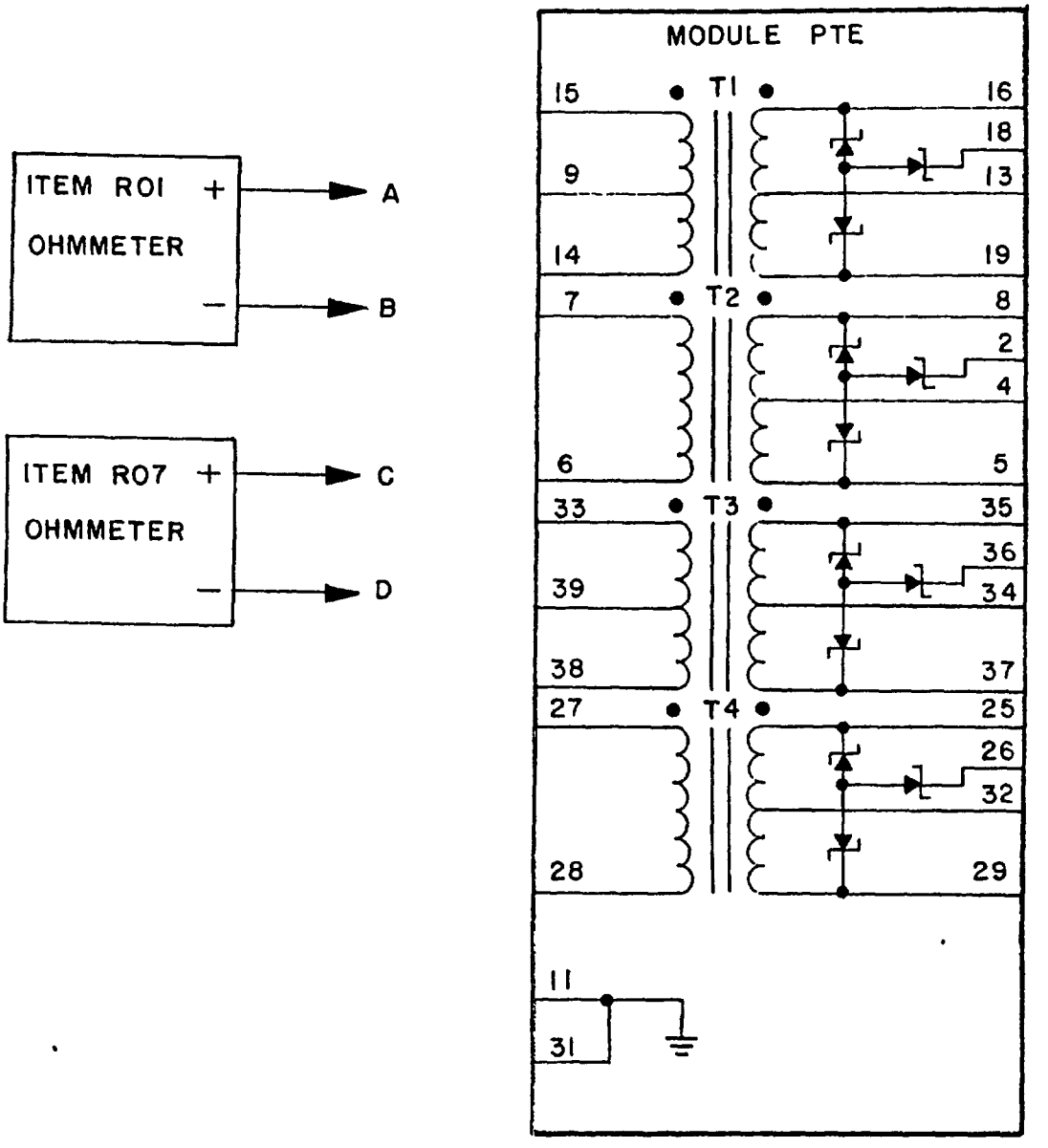


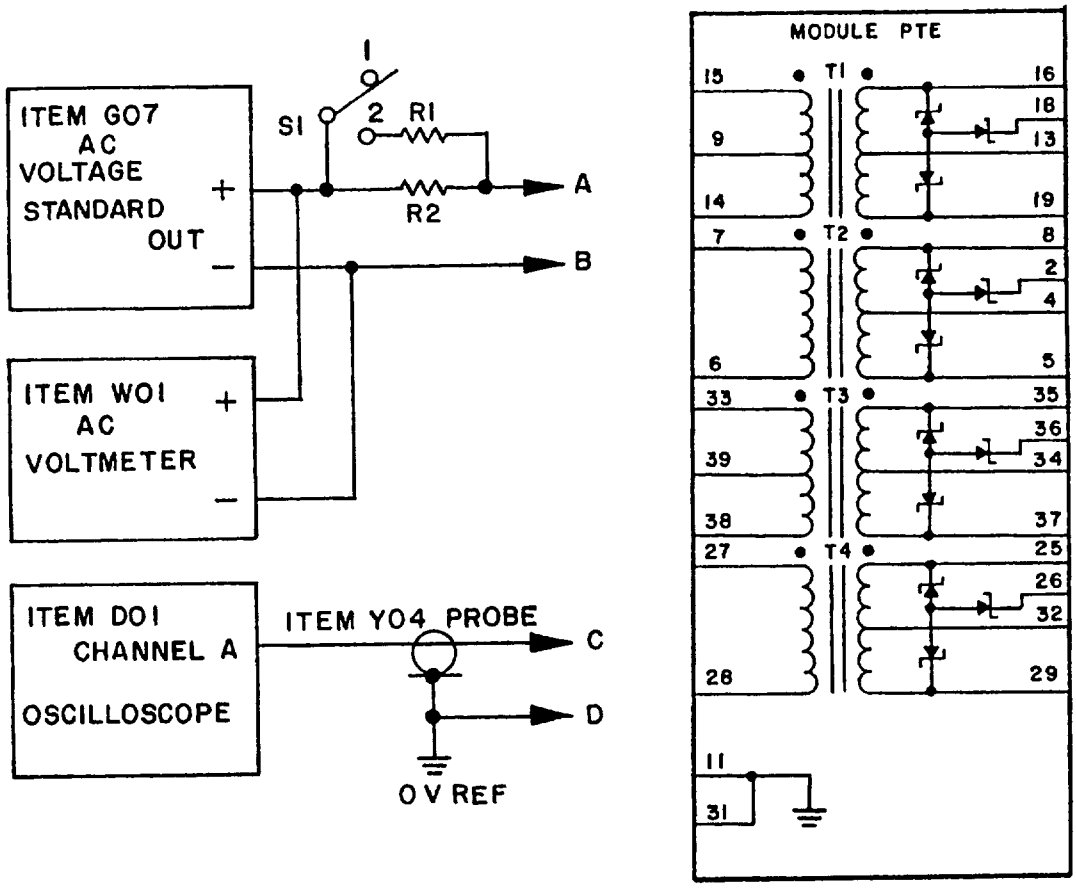
FIGURE 4. Continuity and DC resistance test circuit for tables III and IV.

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TABLE V. Suppression voltage test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
D1	Suppression voltage	18	16,19	18	16,19	16.0	26.0	V_{p-p}	5
D2		2	5,8	2	5,8				
D3		36	35,37	36	35,37				
D4		26	25, 29	26	25,29				
D5		18	16,19	18	16,19				
D6		2	5,8	2	5,8				
D7		36	35,37	36	35,37				
D8		26	25,29	26	25,29				
D9		14	15	16	19				
D10		33	38	35	37				
D11		7	6	8	5				
D12	Suppression voltage	27	28	25	29	16.0	26.0	V_{p-p}	5

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* FIGURE 5. Suppression voltage test circuit for table V.

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NOTES:
1. R1 = 10Ω ±1%, 2 W.
   R2 = 301Ω ±1%, 1/8 W.
   S1 = SPDT
2. Test conditions D1-D4:

   a. Connect the equipment as specified in table V for the appropriate test condition. Set S1 to position 2.
   b. Adjust item G07 for 10.9 ±0.2 Vrms at 2 kHz ±200 Hz at point A.
   c. Set the item D01 vertical sensitivity for 5V/CM.
   d. The peak to peak voltage measured on item D01 shall satisfy the limits of table I.

3. Test conditions D5-D8:

   a. Connect the equipment as specified in table V for the appropriate test condition. Set S1 to position 1.
   b. Perform test per notes 2b, 2c, and 2d.

4. Test conditions D9-D10:

   a. Connect the equipment as specified in table V for the appropriate test condition. Set S1 to position 1.
   b. Adjust item G07 for 230 ±2.0 Vrms at 2 kHz ±200 Hz at point A.
   c. Perform test per notes 2c and 2d.

5. Test conditions D11-D12:

   a. Connect the equipment as specified in table V for the appropriate test condition. Set S1 to position 1.
   b. Adjust item G07 for 200 ±2.0 Vrms at 2 kHz ±200 Hz at point A.
   c. Perform test per notes 2c and 2d.

* FIGURE 5. Suppression voltage test circuit for table V - Continued.

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TABLE VI. Polarity test.

Test condition	Parameter	Connect							Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
E1	Polarity	15	7	14	8	5	16	19	3.2	3.8	V	6
* E2	Polarity	33	27	38	25	29	35	37	3.2	3.8	V	6

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TABLE VII. Angular accuracy test.

Test cond	Parameter	Connect							Set F03 to angle (deg)	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to Pin	D to pin	E to pin	F to pin	G to pin		Min	Max		
F1	Angular accuracy	15	7	14	8	5	16	19	0.000	359.980	0.020	Deg	6
F2									45.000	44.980	45.020		
F3									90.000	89.980	90.020		
F4									20.000	19.960	20.040		
F5									70.000	69.960	70.040		
F6									110.000	109.960	110.040		
F7									160.000	159.960	160.040		
F8									200.000	199.960	200 040		
F9									250.000	249.960	250 040		
F10									290.000	289.960	290.040		
F11		15	7	14	8	5	16	19	340.000	339.960	340.040		
F12		33	27	38	25	29	35	37	0.000	359.980	0.020		
F13									45.000	44.980	45.020		
F14									90.000	89.980	90.020		
F15									20.000	19.960	20.040		
F16									70.000	69.960	70.040		
F17									110.000	109.960	110.040		
F18									160.000	159.960	160.040		
F19									200.000	199.960	200.040		
F20									250.000	249.960	250.040		
F21									290.000	289.960	290.040		
F22	Angular accuracy	33	27	38	25	29	35	37	340.000	339.960	340.040	Deg	6

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TABLE VII. Angular accuracy test -Continued.

Test cond	Parameter	Connect							Set F03 to angle (deg)	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to Pin	D to pin	E to pin	F to pin	G to pin		Min	Max		
F23	Angular accuracy	15	7	14	4	5	13	19	45.000	44.980	45.020	Deg	6
F24									20.000	19.960	20.040		
F25									70.000	69.960	70.040		
F26									110.000	109.960	110.040		
F27									160.000	159.960	160.040		
F28									200.000	199.960	200.040		
F29									250.000	249.960	250.040		
F30									290.000	289.960	290.040		
F31		15	7	14	4	5	13	19	340.000	339.960	340.040		
F32		33	27	38	32	29	34	37	45.000	44.980	45.020		
F33									20.000	19.960	20.040		
F34									70.000	69.960	70.040		
F35									110.000	109.960	110.040		
F36									160.000	159.960	160.040		
F37									200.000	199.960	200.040		
F38									250.000	249.960	250.040		
F39									290.000	289.960	290.040		
F40		33	27	38	32	29	34	37	340.000	339.960	340.040		
F41		15	7	14	8	5	16	19	0.000	359.930	0.070		
F42									45.000	44.930	45.070		
F43									90.000	89.930	90.070		
F44	Angular accuracy	15	7	14	8	5	16	19	20.000	19.930	20.070	Deg	6

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TABLE VII. Angular accuracy test -Continued.

Test cond	Parameter	Connect							Set F03 to angle (deg)	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to Pin	D to pin	E to pin	F to pin	G to pin		Min	Max		
F45	Angular accuracy	15	7	14	8	5	16	19	70.000	69.930	70.070	Deg	6
F46									110.000	109.930	110.070		
F47									160.000	159.930	160.070		
F48									200.000	199.930	200.070		
F49									250.000	249.930	250.070		
F50									290 000	289.930	290.070		
F51		15	7	14	8	5	16	19	340.000	339.930	340.070		
F52		33	27	38	25	29	35	37	0.000	359.930	0.070		
F53									45.000	44.930	45.070		
F54									90.000	89.930	90.070		
F55									20.000	19.930	20.070		
F56									70.000	69.930	70.070		
F57									110.000	109.930	110.070		
F58									160.000	159.930	160.070		
F59									200.000	199.930	200.070		
F60									250.000	249.930	250.070		
F61									290.000	289.930	290.070		
F62		33	27	38	25	29	35	37	340.000	339.930	340.070		
F63		15	7	14	4	5	13	19	45.000	44.930	45.070		
F64		15	7	14	4	5	13	19	20.000	19.930	20.070		
F65		15	7	14	4	5	13	19	70.000	69.930	70.070		
F66	Angular accuracy	15	7	14	4	5	13	19	110.000	109.930	110.070	Deg	6

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TABLE VII. Angular accuracy test -Continued.

Test cond	Parameter	Connect: A to pin	B to pin	C to Pin	D to pin	E to pin	F to pin	G to pin	Set F03 to angle (deg)	Initial limits 25° ±5°C ambient: Min	Max	Unit	Test figure
F67	Angular accuracy	15	7	14	4	5	13	19	160.000	159.930	160.070	Deg	6
F68									200.000	199.930	200.070		
F69									250.000	249.930	250.070		
F70									290.000	289.930	290.070		
F71		15	7	14	4	5	13	19	340.000	339.930	340.070		
F72		33	27	38	32	29	34	37	45.000	44.930	45.070		
F73									20.000	19.930	20.070		
F74									70.000	69.930	70.070		
F75									110.000	109.930	110.070		
F76									160.000	159.930	160.070		
F77									200.000	199.930	200.070		
F78									250.000	249.930	250.070		
F79									290.000	289.930	290.070		
F80	Angular accuracy	33	27	38	32	29	34	37	340.000	339.930	340.070	Deg	6

*Note; Table VII limits (Deg) = angle set on F03 ± table I limits.

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TABLE VIII. Differential quadrature test.

Test cond	Parameter	Connect: A to pin	B to pin	C to Pin	D to pin	E tc pin	F to pin	G to pin	Set F03 to angle (deg)	Initial limits 25° ±5°C ambient: Min	Max	Unit	Test figure
G1	Differential quadrature	15	7	14	8	5	16	19	0.000	-3.5	3.5	mV	6
G2									45.000				
G3									90.000				
G4									20.000				
G5									70.000				
G6									110.000				
G7									160.000				
G8									200.000				
G9									250.000				
G10									290.000				
G11		15	7	14	8	5	16	19	340.000				
G12		33	27	38	25	29	35	37	0.000				
G13									45.000				
G14									90.000				
G15									20.000				
G16									70.000				
G17									110.000				
G18									160.000				
G19									200.000				
G20									250.000				
G21									290.000				
G22	Differential quadrature	33	27	38	25	29	35	37	340.000	-3.5	3.5	mV	6

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TABLE VIII. Differential quadrature test -Continued.

Test cond	Parameter	Connect							Set FO3 to angle (deg)	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to Pin	D to pin	E to pin	F to pin	G to pin		Min	Max		
G23	Differential quadrature	15	7	14	4	5	13	19	45.000	-1.50	1 50	mV	6
G24									20.000				
G25									70.000				
G26									110.000				
G27									160.000				
G28									200.000				
G29									250.000				
G30									290.000				
G31		15	7	14	4	5	13	19	340.000				
G32		33	27	38	32	29	34	37	45.000				
G33									20.000				
G34									70.000				
G35									110.000				
G36									160.000				
G37									200.000				
G38									250.000				
G39									290.999				
G40		33	27	38	32	29	34	37	340.000	-1.50	1.50		
G41		15	7	14	8	5	16	19	0.000	-3.75	3.75		
G42		15	7	14	8	5	16	19	45.000	-3.75	3.75		
G43	Differential quadrature	15	7	14	8	5	16	19	90.000	-3.75	3.75	mV	6

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TABLE VIII. Differential quadrature test -Continued

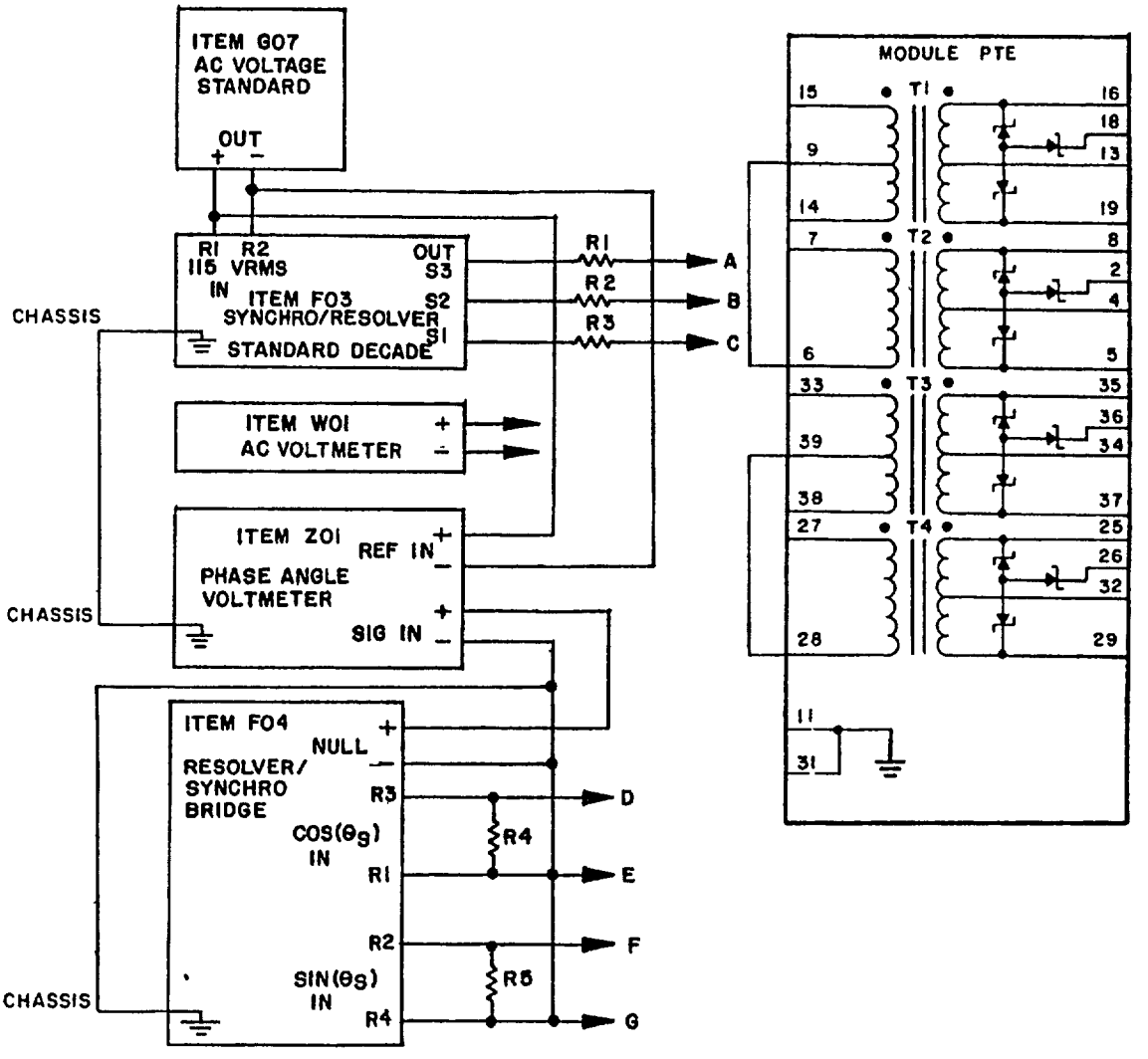
Test cond	Parameter	Connect: A to pin	B to pin	C to Pin	D to pin	E to pin	F to pin	G to pin	Set F03 to angle (deg)	Initial limits 25° ±5°C ambient: Min	Max	Unit	Test figure
G44	Differential quadrature	15	7	14	8	5	16	19	20.000	-3.75	3.75	mV	6
G45									70.000				
G46									110.000				
G47									160.000				
G48									200.000				
G49									250.000				
G50									290.000				
G51		15	6	14	8	5	16	19	340.000				
G52		33	27	38	25	29	35	37	0.000				
G53									45.000				
G54									90.000				
G55									20.000				
G56									70.000				
G57									110.000				
G58									160.000				
G59									200.000				
G60									250.000				
G61									290.000				
G62		33	27	38	25	29	35	37	340.000	-3.75	3.75		
G63		15	7	14	4	5	13	19	45.000	-1.75	1.75		
G64	Differential quadrature	15	7	14	4	5	13	19	20.000	-1.75	1.75	mV	6

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TABLE VIII. Differential quadrature test -Continued.

Test cond	Parameter	Connect							Set FO3 to angle (deg)	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to Pin	D to pin	E to pin	F to pin	G to pin		Min	Max		
G65	Differential quadrature	15	7	14	4	5	13	19	70.000	-1.75	1.75	mV	6
G66									110.000				
G67									160.000				
G68									200.000				
G69									250.000				
G70									290.000				
G71		15	7	14	4	5	13	19	340 000				
G72		33	27	38	32	29	34	37	45.000				
G73									20.000				
G74									70.000				
G75									110.000				
G76									160.000				
G77									200.000				
G78									250.000				
G79									290.000				
G80	Differential quadrature	33	27	38	32	29	34	37	340.000	-1.75	1.75	mV	6

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NOTE: For F03 $E_{(S1-S3)} = 0.783E_{(R2-R1)} \text{ SIN } (\theta_S)$

$E_{(S3-S2)} = 0.783E_{(R2-R1)} \text{ SIN } (\theta_S + 120^\circ)$

$E_{(S2-S1)} = 0.783E_{(R2-R1)} \text{ SIN } (\theta_S + 240^\circ)$

For F04 $E_{(H1-LO)} = E_{(R3-R1)} \text{ COS } (\theta_B) + E_{(R4-R2)} \text{ SIN } (\theta_B)$

θ_S = Standard angle

θ_B = Bridge angle

FIGURE 6. Polarity test, angular accuracy test and differential quadrature test circuit for tables VI, VII and VIII.

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NOTES:
1. R1, R2, R3 = 23.7Ω ±1%, 1/8 W.
   R4, R5 = 50 kΩ ±1%, 1/8 W.
2. Test condition E:

   a. Connect the equipment as specified in table VI.
   b. With item G07 set to zero, set the angle on item F03 to 90.000°.
   c. Adjust item G07 for 115.0 ±1.1 Vrms at 400 ±4 Hz.
   d. Set item F04 to 0.000°. Set the item Z01 function switch to 0°. The voltage read on item Z01 shall satisfy the limits of table VI. The needle of item Z01 shall deflect to the right to indicate an in phase measurement.

3. Test conditions F1-F40 and G1-G40 (performed concurrently):

   a. Connect the equipment as specified in table VII for the appropriate test condition.
   b. For test conditions F1-F80 and G1-G80 various combinations of switched angles on item F03 may result in magnetic core saturation of the module transformers. To prevent this the output voltage of item G07 shall be linearly reduced to zero volts before angle settings on item F03 are changed. Adjust item F03 to the angle specified for the appropriate test condition of table VII.
   c. Adjust item G07 for 115.0 ±1.1 Vrms at 400 ±4 Hz. Set the item Z01 function switch to 0° and adjust item F04 for a null reading on item Z01.
   d. The setting of item F04 shall satisfy the limits of table VII test conditions.
   e. With no change to the above test set-up (3a-3d), set the function switch of item Z01 to 90°. The quadrature voltage read on item Z01 shall satisfy the limits of table VIII.

4. Test conditions F41-F80 and G41-G80 (performed concurrently):

   a. Perform test set-up per note 3a.
   b. Repeat notes 3b-3e except the frequency of item G07 shall be 800 ±8 Hz.

* FIGURE 6. Polarity test, angular accuracy test and differential quadrature test circuit for tables VI, VII and VIII - Continued.

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TABLE IX. Common mode rejection ratio test.

Test condition	Parameter	Connect					Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min 1/	Max 1/		
H1	Common mode rejection ratio	7	14	15	16	19	0	2	mV p-p	7
H2		7	14	15	8	5		2		
H3		27	33	38	35	37		2		
H4		27	33	38	25	29		2		
H5		7	14	15	16	19		40		
H6		7	14	15	8	5		40		
H7		27	33	38	35	37		40		
H8	Common mode rejection ratio	27	33	38	25	29	0	40	mV p-p	7

1/Table IX limits (volts) = 100 $\left[10^{(X/20)}\right]$ where X = table I limits (dB)

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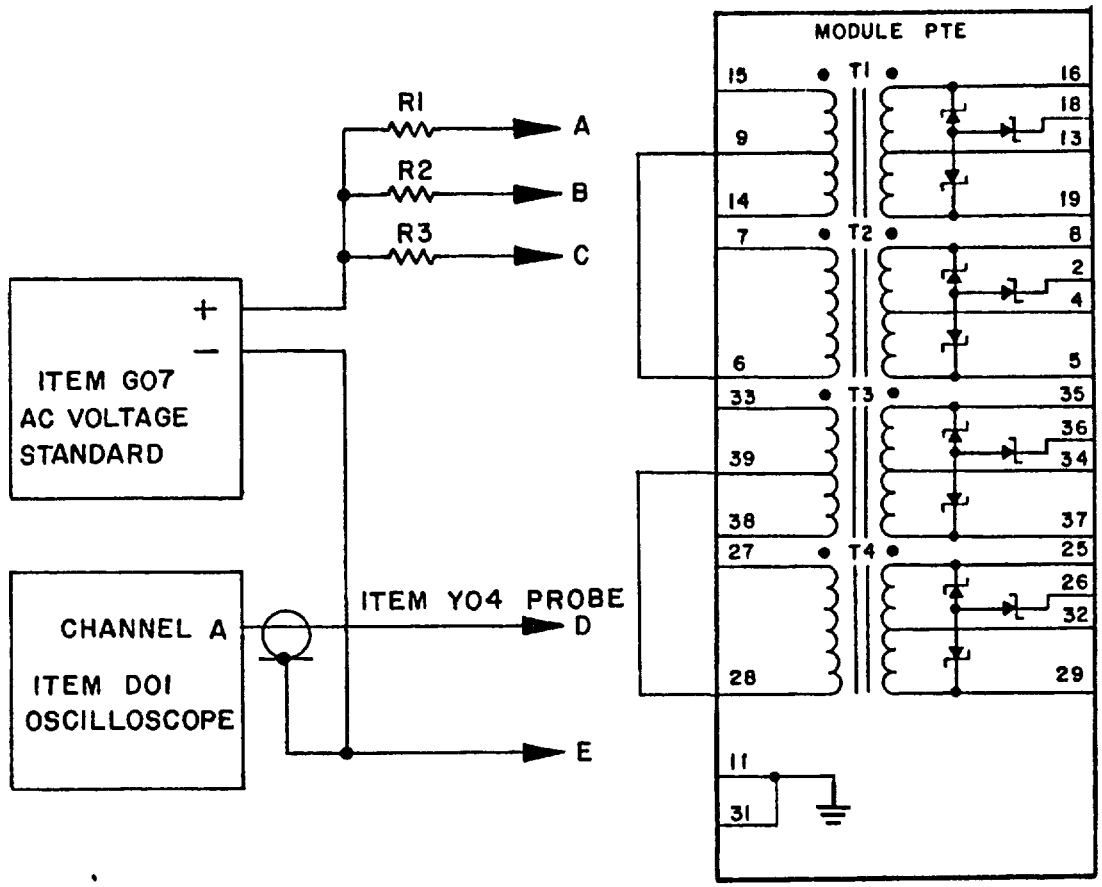


FIGURE 7. Common mode rejection ratio test circuit for table IX.

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NOTES:
1. R1, R2, R3 = 23.7Ω ±1%, 1/8 W.
2. Test conditions H1-H4:

   a. Connect the equipment as specified in table IX for the appropriate test condition.
   b. Adjust item G07 for 100 ±1 volts peak-to-peak at 1 ±0.01 kHz.
   c. The peak-to-peak voltage measured on item D01 shall satisfy the limits of table IX.

3. Test condition H5-H8:

   a. Perform tests per notes 2a, 2b, and 2c with the frequency of item G07 set for 10 ±0.1 kHz.

4. The requirement (R) of table I and the measured voltage of table IX have the following relationship:

$$R_{(dB)} = 20 \text{ Log } \left[\frac{V \text{ (Measured) (Volts)}}{100 \quad \text{(Volts)}} \right]$$

FIGURE 7. Common mode rejection ratio test circuit for table IX - Continued.

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TABLE X. Transformation ratio test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
J1	Transformation ratio	15	14	16	19	.03850	.03928	-	8
J2	↓	33	38	35	37	.03850	.03928	↓	↓
J3	↓	15	14	13	19	.01100	.01122	↓	↓
J4	Transformation ratio	33	38	34	37	.01100	.01122	-	8

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TABLE XI. Phase shift test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min 1/	Max 1/		
K1	Phase shift	15	14	16	19	-6.0	6.0	mV	8
K2	↓	33	38	35	37	-6.0	6.0	↓	↓
K3	↓	15	14	16	19	-12.0	12.0	↓	↓
K4	Phase shift	33	38	35	37	-12.0	12.0	mV	8

1/ Table XI limits (volts) = 3.5 tan [table I limits (deg)]

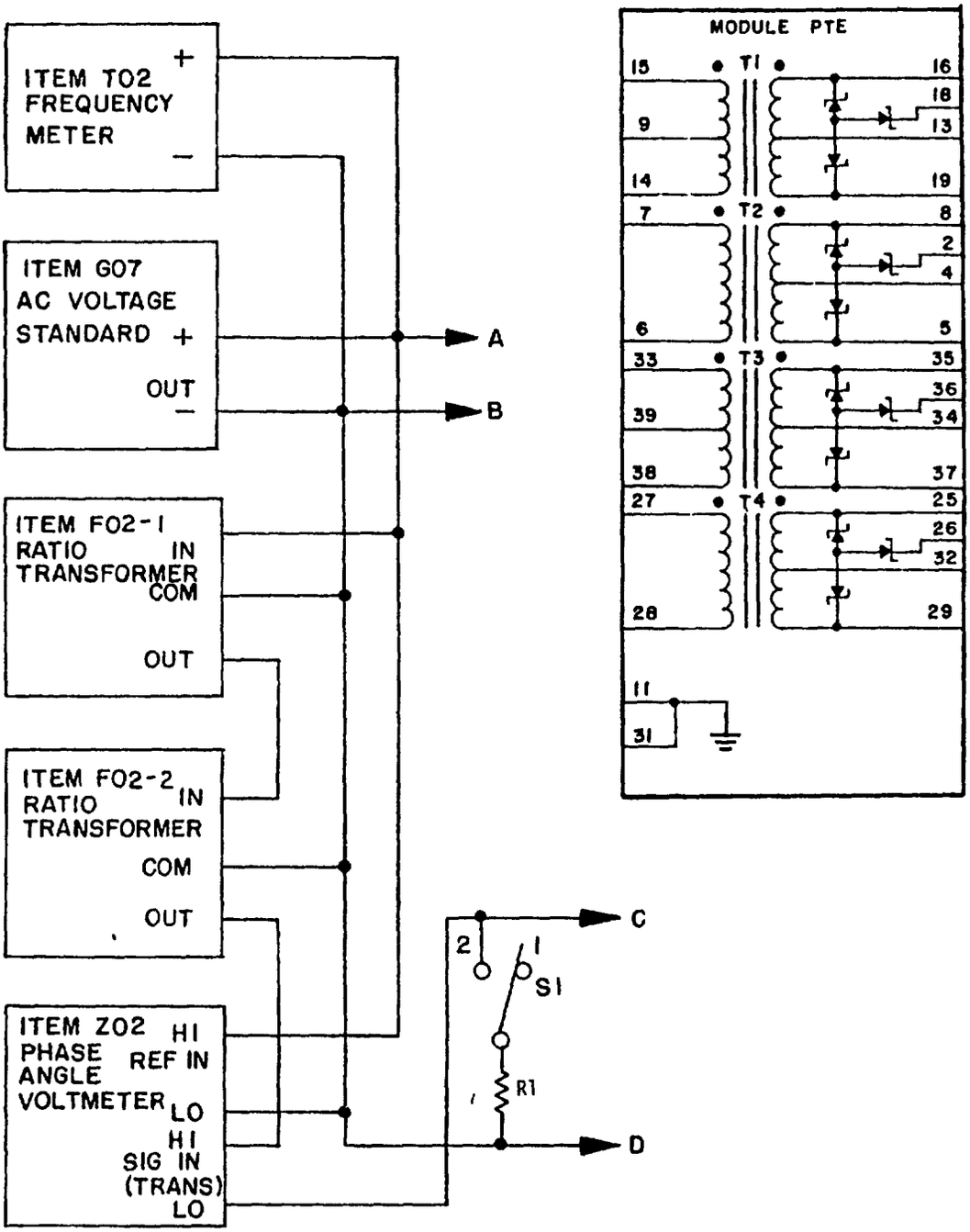
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TABLE XII. Load regulation test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min 1/	Max 1/		
L1	Load regulation	15	14	16	19	.999	.99995	-	8
L2	↓	7	6	8	5	.999	.99995	↓	↓
L3	↓	33	38	35	37	.999	.99995	↓	↓
L4	Load regulation	27	28	25	29	.999	.99995	-	8

1/ Table XII limits = $\left[1 - \frac{(\text{table I limits})(1\%)}{100}\right]$

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* FIGURE 8. Transformation ratio test, phase shift test and load regulation test circuit for tables X, XI and XII.

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NOTES:
1. S1 = SPDT switch.
   R1 = 50 kΩ ±1%, 1/8 W.
2. For test conditions J1-J4 and K1-K4, item F02-2 shall be set to 1.000000 and S1 shall be in position 1.
3. Test conditions J1-J4:

   a. Connect the equipment as specified in table X for the appropriate test condition.
   b. Adjust item G07 for 90 ±1.0 Vrms at 400 ±4 Hz at point A.
   c. With the function switch of item Z02 set to 0°, adjust item F02-1 for a null reading on item Z02.
   d. The setting of item F02-1 shall satisfy the limits of table X.

4. Test condition K1 and K2:

   a. Connect the equipment as specified in table XI for the appropriate test condition.
   b. Repeat procedure of notes 3b and 3c.
   c. With no change to the set up, set the function switch of item Z02 to 90°. The quadrature voltage read on item Z02 shall satisfy the limits of table XI.

5. Test condition K3 and K4:

   a. Connect the equipment as specified in table XI.
   b. Adjust item G07 for 90 ±1.0 Vrms at 800 ±8 Hz at point A.
   c. Repeat procedure of note 3c.
   d. Switch the function switch of item Z02 to 90°. The quadrature voltage read on item Z02 shall satisfy the limits of table XI.

6. The requirement (R) of table I and the voltage measured in test conditions K1 through K4 have the following relationship:

$$R_{(DEG)} = \text{Arctan} \left[\frac{V(\text{Measured}) \quad (\text{Volts})}{3.5 \quad (\text{Volts})} \right]$$

7. Test conditions L1 and L3:

   a. Connect the equipment as specified in table XII for the appropriate test condition with S1 in position 1.
   b. Repeat procedure of note 3b and 3c.
   c. Set S1 to position 2 and adjust item F02-2 for a null reading on item Z02.
   d. The setting of item F02-2 shall satisfy the limits of table XII.

8. Test conditions L2 and L4:

   a. Repeat procedure of note 7a.
   b. Adjust item G07 for 78 ±1 Vrms at 400 ±4 Hz at point A.
   c. Repeat procedure of note 3c, 7c and 7d.

9. The requirement (R) of table I and the setting of ratio transformer have the following relationship:

$$R\ (\%) = (100)\ (1\text{-}RT)$$

RT = Item F02-2 setting for the 0° null reading.

* FIGURE 8. Transformation ratio test, phase shift test and load regulation test circuit for tables X, XI and XII - Continued.

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* TABLE XIII. Input impedance test, input impedance balance test and input impedance angle test.

Test condition	Parameter	Connect			Set F03 to angle (deg)	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin		Min	Max		
M1	Input impedance	15	7	14	30.000	30 1/	90 1/	mV	9
M2	↓	15	7	14	90.000	30 1/	90 1/	mV	↓
M3	↓	33	27	38	30.000	30 1/	90 1/	mV	↓
M4	Input impedance	33	27	38	90.000	30 1/	90 1/	mV	↓
M5	Input impedance balance	-	-	-	- -	0	20	%	↓
M6	Input impedance balance	-	-	-	-	0	20	%	↓
M7	Input impedance angle	15	7	14	30.000	5 2/	85 2/	Deg	↓
M8	↓	15	7	14	90.000	5 2/	85 2/	Deg	↓
M9	↓	33	27	38	30.000	5 2/	85 2/	Deg	↓
M10	Input impedance angle	33	27	38	90.000	5 2/	85 2/	Deg	↓
M11	Input impedance	15	7	14	30.000	20 1/	90 1/	mV	↓
M12	↓	15	7	14	90.000	20 1/	90 1/	mV	↓
M13	↓	33	27	38	30.000	20 1/	90 1/	mV	↓
M14	Input impedance	33	27	38	90.000	20 1/	90 1/	mV	↓
M15	Input impedance angle	15	7	14	30.000	40 2/	85 2/	Deg	↓
M16	↓	15	7	14	90.000	40 2/	85 2/	Deg	↓
M17	↓	33	27	38	30.000	40 2/	85 2/	Deg	↓
M18	Input impedance angle	33	27	38	90.000	40 2/	85 2/	Deg	9

1/ Table XIII limits (volts) = $\frac{\text{9000 (Volt-ohms)}}{\text{table I limits (ohms)}}$

2/ Table XIII limits (Deg) = 90° - table I limits (Deg)

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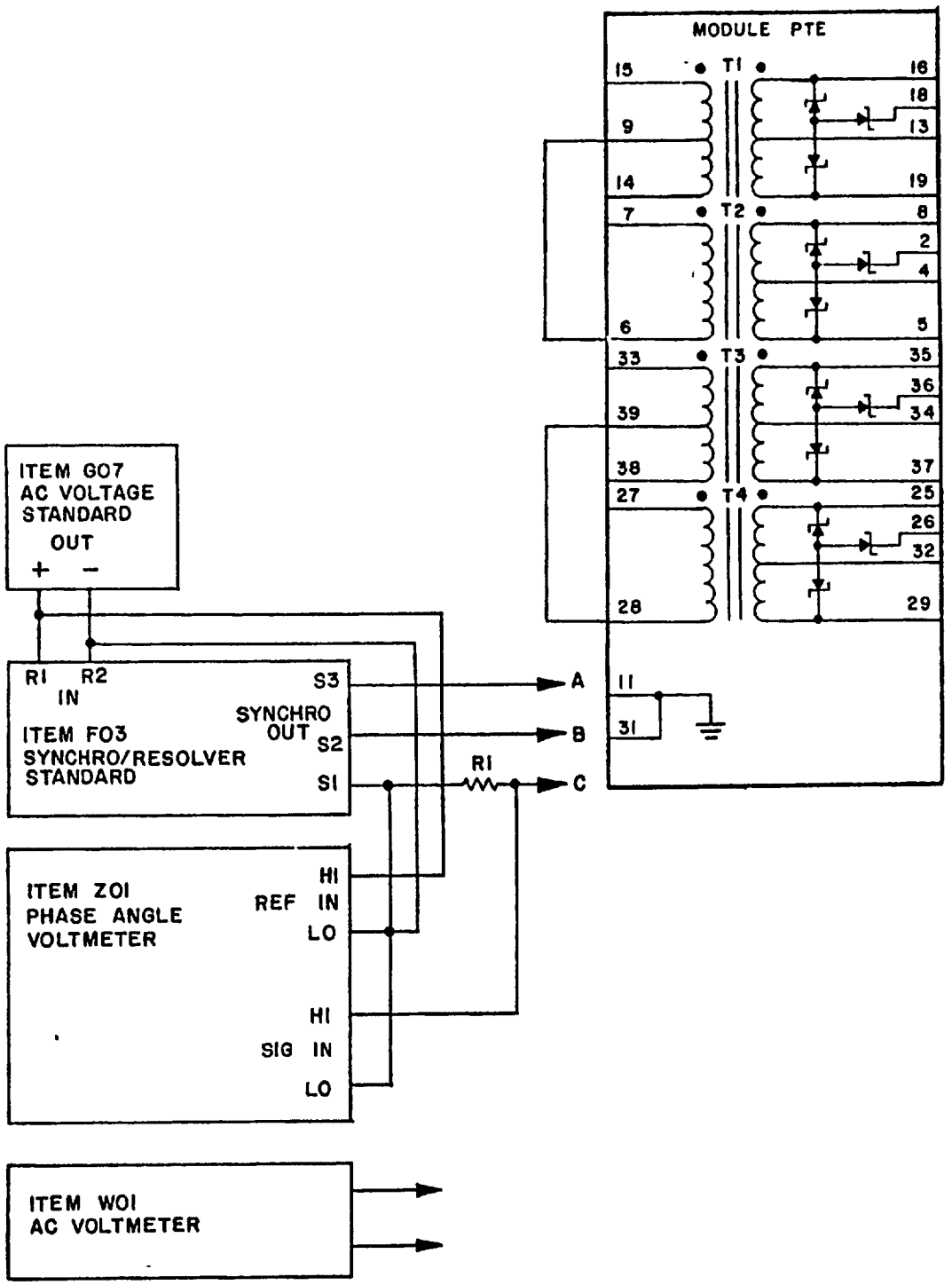


FIGURE 9. Input impedance test, input impedance balance test and input impedance angle test circuit for table XIII.

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NOTES:
1. R1 = 100Ω ±1%, 1/8 W.
2. Test conditions M1-M4:

   a. Connect the equipment as specified in table XIII for the appropriate test condition.
   b. Adjust items G07 and F03 for a 90 ±1.0 Vrms at 400 ±4 Hz synchro signal measured between synchro outputs S1 and S3 of item F03 with the angle set at 90.000°.
   c. With the function switch of item Z01 in the fund position, measure and record the voltage reading on item Z01. The reading shall satisfy the limits of table XIII.

3. Test conditions M5-M6:

   a. For test condition M5 requirement (R) of table XIII:

$$R\ (\%) = \frac{A-B}{A} \times 100$$

   A = Larger voltage recorded for note 2c for test conditions M1 and M2.
   B = Smaller voltage recorded for note 2c for test conditions M1 and M2.

   b. For test condition M6, perform the test per note 3a for voltages recorded for test conditions M3 and M4.

4. Test conditions M7-M10:

   a. Repeat procedure of notes 2a and 2b.
   b. With the function switch of item Z01 in the 0° position, adjust the degrees dial for a null indication. The reading on the degrees dial shall satisfy the limits of table XIII. Set function switch to 90°, needle shall deflect to left.

5. Test conditions M11-M14:

   a. Connect the equipment as specified in table XIII for the appropriate test conditions.
   b. Adjust items G07 and F03 for a 90 ±1.0 Vrms at 800 ±8 Hz synchro signal measured between synchro outputs S1 and S3 of item F03 with the angle set at 90.000°.
   c. Perform tests per note 2c.

6. Test conditions M15-M18:

   a. Repeat procedure of notes 5a and 5b.
   b. Perform tests per note 4b.

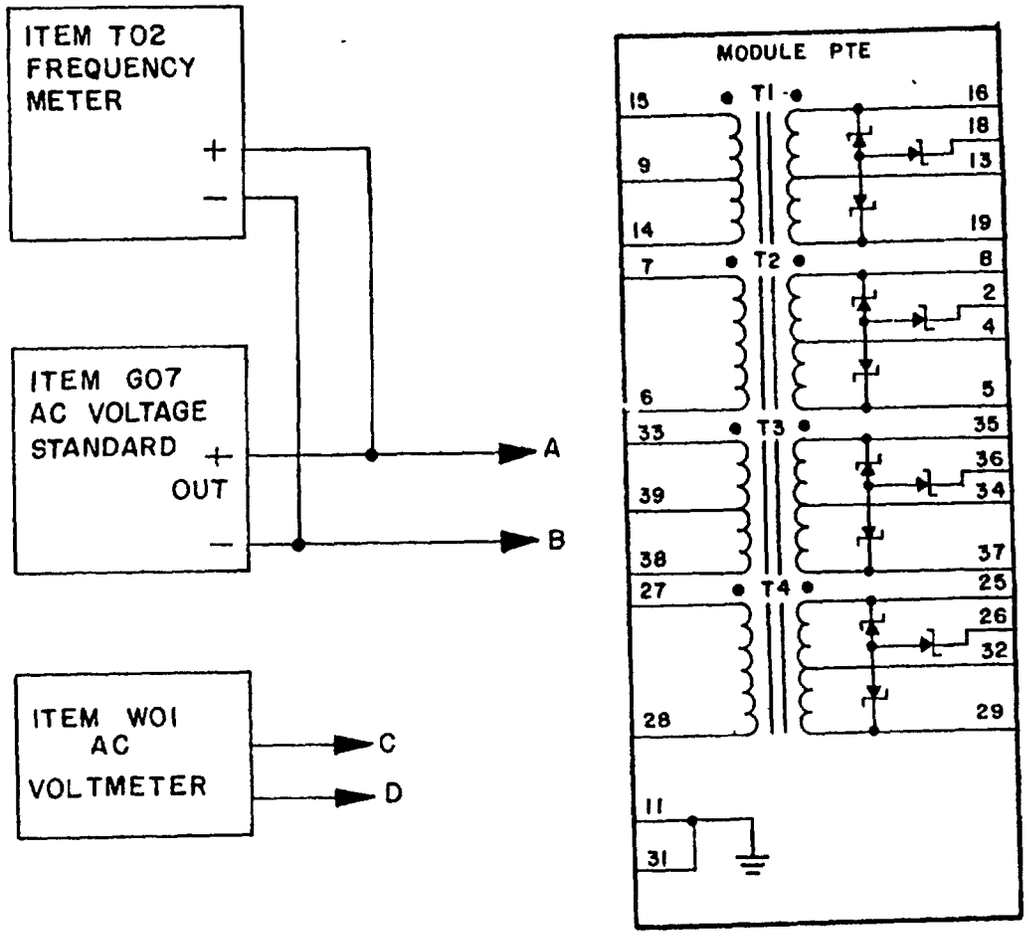
FIGURE 9. Input impedance test, input impedance balance test and input impedance angle test circuit for table XIII - Continued.

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TABLE XIV. Cross talk test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	Crosstalk	15	14	35	37	0.0	0.2	mV	10
N2		15	14	25	29				
N3		15	14	8	5				
N4		33	38	16	19				
N5		33	38	8	5				
N6		33	38	25	29				
N7		7	6	35	37				
N8		7	6	25	29				
N9		7	6	16	19				
N10		27	28	16	19				
N11		27	28	8	5				
N12	Crosstalk	27	28	35	37	0.0	0.2	mV	10

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NOTES:

1. Test conditions N1-N6:

   a. Connect the equipment as specified in table XIV for the appropriate test condition.
   b. Adjust item G07 for 90 ±1.0 Vrms at 800 ±8 Hz at point A.
   c. Measure the voltage on item W01. The measured value shall meet the limits of table XIV.

2. Test conditions N7-N12:

   a. Perform tests per notes 2a, 2b, and 2c with the output voltage amplitude of item A01 adjusted to 78 ±1.0 Vrms.

* FIGURE 10. Crosstalk test circuit for table XIV.

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